

Pivacidade de Sistemas

Systems Privacy

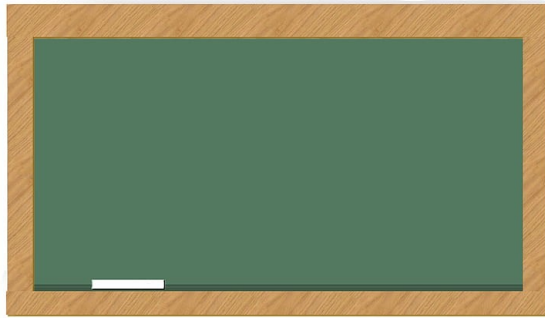
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Today's class will introduce *privacy* and *this semester's outline and material*.

I will be going through different *definitions of privacy*, as well as their *implications* for developers. This includes *contextual integrity*, *Solove's taxonomy*, and more.

This is a *complex* topic, and knowledge about privacy comes from *multiple disciplines*. For this week we will be covering understandings from *philosophy*, *law*, and *computer science*.

Who Am I



- Systems Privacy
- Reliable Distributed Systems
- Socio-professional Aspects of Informatics
- ...



- Access Control
- Privacy
- Anonymity
- Usable Security and Privacy
- ...

Who Am I

Privacy | Lx



PdS's Lesson Structure

- Definitions, Law, and Philosophy (3)
 - Defining Privacy
 - Legal Privacy Protections
 - GDPR
 - CCPA
 - Privacy and Ethics
- Web and Data Privacy (3)
 - Web Tracking
 - Web Privacy
 - De-identification
 - re-identification
 - K-Anonymity
 - T-diversity
 - I-closeness
 - Differential Privacy
 - Local differential privacy
- Anonymity (2)
 - Mix nets
 - DC Nets
 - Tor
- PET Building Blocks (4)
 - Multiparty Computation
 - Zero-knowledge Proofs
 - Private Information Retrieval
 - ...

PdS's Evaluation Structure

- Tests (55%)
 - Two tests
 - Mid-semester
 - End of semester
 - Average grade of tests is worth 50% of the final grade
 - No consultation
 - 2 hour duration
- Projects (45%)
 - Two projects throughout the semester
 - One on data privacy, one on anonymity
 - To be done in groups of two
 - I can accept individuals working alone, but please consider doing it with a partner
 - Short technical report per project
 - Brevity is a skill, and more pages doesn't mean better quality
 - Due dates TBD

Exact dates TBD

PdS's Evaluation Structure

- Exam
 - One exam at the end of the semester for those who:
 - Have **frequency**; and
 - Wish to **improve** their grades.
 - This exam will be in the same format as the first two, but will cover **all class material**.
- Frequency:
 - In order to have frequency, you must:
 - Have an **average of 8,0** or better on your projects
- Grading guidelines for the projects will be released at the same time as the projects.
- All grading information can be found on CLIP.

Readings

- This class will depend on a lot of **academic readings**.
 - Understanding academic papers is a skill needed for a thesis.
- I **may** ask about readings on the tests.
- Readings should be done **before the class** if possible.
- Readings can always be found on the **slides** and on **CLIP**.

Questions?



Privacy

Why Privacy?

Imagine a person who committed a crime in their 20s, but in their 40s have become a respected and contributing member of society. Should information about their past crime come to light, how do you think the community would act towards them?

*Imagine a **teenager** realizing they **don't believe in their parents' religion**, but their parents are **religious extremists**. What could happen if the child's lack of faith is **discovered**?*

*Imagine a person **expressing their thoughts** on a topic **before being 100% informed** on it. Should their old opinions **forever be archived and accessible** by others?*

Why Privacy?

- Privacy is a **human right**, recognized by various countries and the international community
- Privacy is an **enabling right**, which helps people:
 - **Form opinions** free from coercion
 - **Express themselves** more deeply
 - Form more **intimate relationships**
 - **Recover** from past mistakes
- Privacy is becoming more **difficult to achieve** given the monetization strategy of modern Internet companies.

A* Privacy Problem

* One of many

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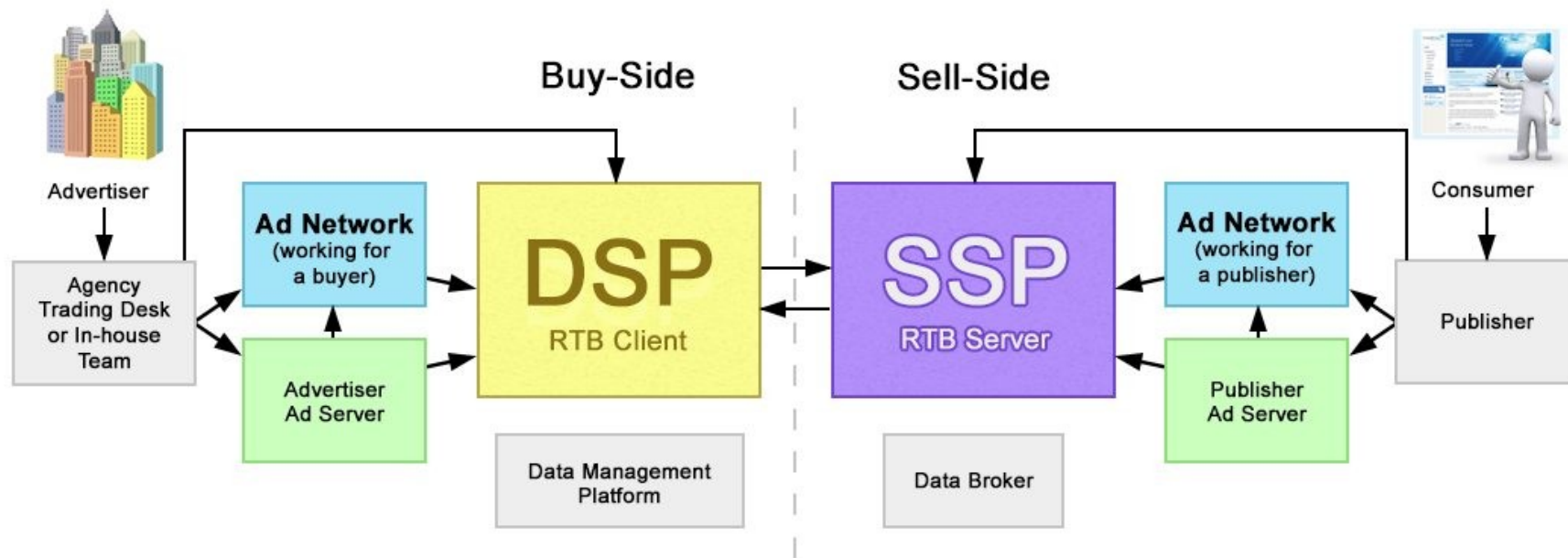
*You search for a **set of boots** online.*

*You decide not to buy them. They're **too expensive**.*

*Most of the **ads you see** on the next pages are **all about those boots**.*

*The current Internet is **not** designed to have privacy.*

Advertising on the Internet



*Before we can understand **how privacy is protected**, we first need to know **what privacy is**.*

What is Privacy?

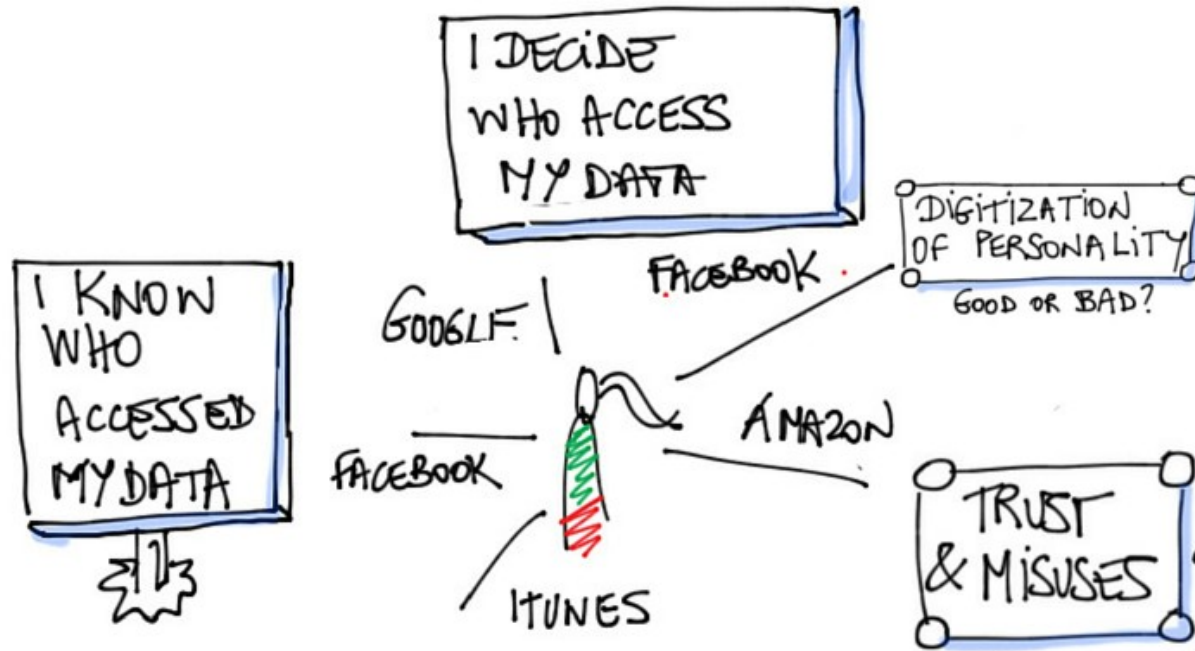
Privacy as Secrecy



Privacy as Secrecy

- **Simplest** definition of privacy.
- Information is kept **private** if it is kept **completely secret**.
 - The moment information is revealed to **any other person**, it loses its private status.
- Pros:
 - **Simplicity**
 - Easy **application** of the definition
- Cons:
 - Over simplistic, **missing most desired privacy guarantees**
- **Legal precedent*** for this definition
 - **Third Party Doctrine** - US

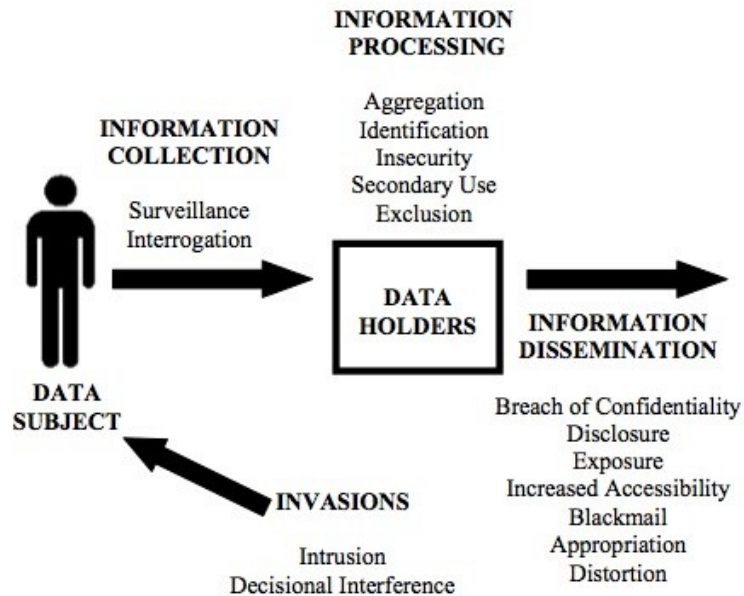
Privacy as Autonomy Over Data



Privacy as Autonomy Over Data

- **Simple** definition of privacy.
- Information is considered **private** if the data subject **considers it private**.
 - It cannot be used in any way the data subject **objects** to.
- Pros:
 - **Simplicity**
 - Strong privacy protections
- Cons:
 - Over simplistic, **missing some cases in which privacy isn't desired**
- Forms the basis of **some** current privacy guarantees.

Solove's Taxonomy



Solove's Taxonomy

- **Legal** work by Daniel J. Solove from 2005.
- Taxonomy of **concrete privacy harms**
 - Focus on the **activities** that invade privacy, **not a definition**
- Privacy is a general umbrella term.
 - Privacy reduced to an “**injury to feelings**”
- “Privacy is the relief from a **range** of kinds of **social friction**”
- Privacy issues have a “complicated **network of similarities** overlapping and criss-crossing”
- Goals are to **identify**:
 - **How** these privacy problems **diverge** from each other
 - **How** these privacy problems **relate** to each other

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Solove: Information Collection

- Harm 1: Surveillance
 - Watching, listening to, or recording activities
- Harm 2: Interrogation
 - Questioning or probing for information

Solove: Information Processing

- Harm 1: **Aggregation**
 - Combination of **multiple pieces of data** about a subject
 - This includes **inferring** new pieces of information based on the info you have
 - i.e. prediction or inference from artificial intelligence or other methods
- Harm 2: **Identification**
 - **Linking** information to a specific **individual**
- Harm 3: **Insecurity**
 - **Carelessness** in protecting the stored information

Solove: Information Processing

- Harm 4: **Secondary Use**
 - Use of information collected for **one** purpose for **another purpose**.
- Harm 5: **Exclusion**
 - **Failure** to:
 - **Notify** the data subject of the presence of data, and
 - Allow the data subject's **participation** in its handling and use

Solove: Information Dissemination

- Harm 1: **Breach of Confidentiality**
 - Breaking a promise to keep collected information **confidential**
- Harm 2: **Disclosure**
 - **Revelation** of **truthful information** that impacts the way a person is judged
- Harm 3: **Exposure**
 - Revealing a persona's nudity, grief, or bodily functions
- Harm 4: **Increased Accessibility**
 - Increasing the accessibility of third parties to a person's information

Solove: Information Dissemination

- Harm 5: **Blackmail**
 - Threatening to disclose personal information
- Harm 6: **Appropriation**
 - Use of the data subject's **identity** to serve the aims and interests of another
- Harm 7: **Distortion**
 - Dissemination of **false** or **misleading** information

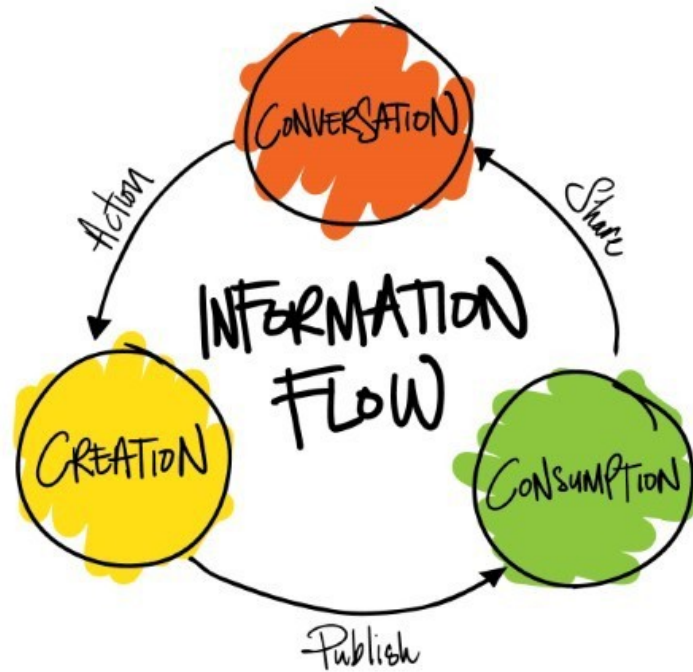
Solove: Invasions

- Harm 1: **Intrusion**
 - Invasive **acts** that **disturb** one's tranquility or solitude
- Harm 2: **Decisional Interference**
 - **Government's** intrusion into the data subject's **decisions** regarding their **private affairs**

Solove's Taxonomy

- Pros:
 - A list of **concrete** privacy harms.
 - More **complex** than previous definitions
 - Still somewhat **easy to apply**
- Cons
 - Doesn't attempt to **define** privacy
 - Some harms are dependent on knowing the **outcome** of an action
 - :For example, see disclosure
 - Some harms are **overly restrictive** in their definition
 - For example, see Decisional Interference

Privacy as Contextual Integrity



Privacy as Contextual Integrity

Does the **Information Flow** according to **Social Norms**?

Privacy as Contextual Integrity

- Appropriate **flow** determined by:
 - Data **Sender**
 - Data **Subject**
 - Data **Recipient**
 - Information **Type**
 - Medical, salary, religious, immigration status, etc.
 - **Transmission Principle**
 - Rules governing how information should or should not flow.
- Has some applicability to the GDPR

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Me
 - Data **Subject**: You
 - Data **Recipient**: You
 - Information **Type**: Grades
 - **Transmission Principle**: In a private email
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Me
 - Data **Subject**: You
 - Data **Recipient**: The whole class
 - Information **Type**: Grades
 - **Transmission Principle**: In a grade publication
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Me
 - Data **Subject**: You
 - Data **Recipient**: A cybersecurity company
 - Information **Type**: Grades, official opinion
 - **Transmission Principle**: With consent; In a recommendation letter
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Me
 - Data **Subject**: You
 - Data **Recipient**: A cybersecurity company
 - Information **Type**: Grades, official opinion
 - **Transmission Principle**: Without consent; in a cold email
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Your doctor
 - Data **Subject**: You
 - Data **Recipient**: Your boss
 - Information **Type**: Health information
 - **Transmission Principle**: In a secure, end-to-end encrypted email
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Your doctor
 - Data **Subject**: You
 - Data **Recipient**: Your boss
 - Information **Type**: Health information about sick leave
 - **Transmission Principle**: In a secure, end-to-end encrypted email
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Example:
 - Data **Sender**: Your computer
 - Data **Subject**: You
 - Data **Recipient**: Your employer
 - Information **Type**: Keystroke data
 - **Transmission Principle**: For the stated purpose of security monitoring
- Does this flow of information violate social norms?

Privacy as Contextual Integrity

- Problems?

Privacy as Contextual Integrity

- Problems:
 - Who decides social norms?
 - Are social norms the same everywhere?
 - How can we enforce contextual integrity?
 - Doesn't it imply we need meta-data for every communication flow?
 - Doesn't this violate social norms?
 - Boils down to the information flow control problem.
- My conclusion:
 - Useful to think about privacy
 - Maybe applicable in some scenarios
 - Mostly can't be enforced.

Any Questions?

References

[1] <https://heinonline.org/HOL/LandingPage?handle=hein.journals/pnlr154&div=20>

[2] <https://heinonline.org/HOL/LandingPage?handle=hein.journals/washlr79&div=16>